1. Introduction

The proposed project was to develop a secure role-based web application that allows students to make submissions for a magazine. To bring brand awareness of the magazine, the name that was chosen was Swift. Although several different technologies could have been used to meet the requirements set, the web application was chosen to be developed purely in PHP, HTML, CSS and using bootstrap as a responsive web design framework for compatibility concerns. The chosen methodology that was used to implement the web application was scrum, which consisted of five team members, each with individual skills and qualities. The roles consisted of a product owner, scrum master and a team ranging from tester, web designer, database engineer and a programmer. This report will provide a detail evaluation on the product, process, team and individual contribution to the project.

Figure 1. Home page (index.php)

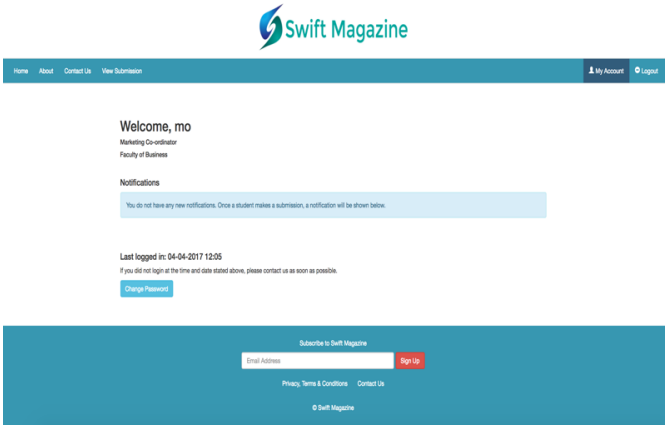

Figure 2. Marketing Co-Ordinator Account

Figure 3. View article online in PDF

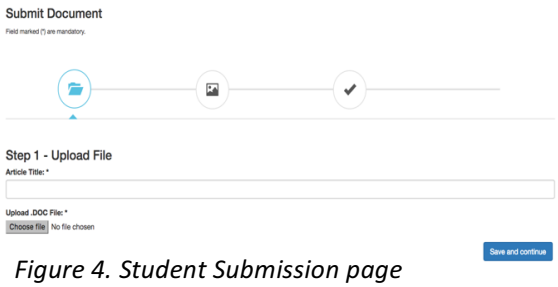

Figure 4. Student Submission page

2. Evaluation

2.1 Project Evaluation

The project objective was to build a secure role-based web application for collecting student contributions for an annual university magazine. The scope of the project was reasonably large; therefore, it was reasonable to work in a scrum team to meet the requirements for the project. Some requirements that were set were not specific, therefore several assumptions were made to benefit the magazine submission website.

The requirements were achieved and developed at the high standard that was required. Firstly, there was on-going communication between the team members and the client to make sure the requirements gathered are specific, clear and understandable to the scrum team. Although, the requirements were not achieved in the first instance, many adjustments were made to the functionalities of the website to achieve the highest client satisfaction by working in an iterative fashion in a scrum team. To meet the requirements, it involved working incrementally therefore each requirement or functionality of the website was developed at the highest standard as possible before moving onto the next functionality.

"All new contributions are disabled after a closure date for new entries, but updates can continue to be done until a final closure date."

An example of working in an iterative fashion, was a functionality that disables the submission year when the deadline has passed. In the first implementation, the administrator had an option to 'Open' or 'Close' the submission year which had to be done manually, therefore if the deadline has passed and the status is still set to 'Open' then it will still allow submission to be made. Although the requirement was still met, as it allowed the admin to open or close the submission year, this may cause issues therefore, the team opted for an automation approach to this functionality. In revision to the first implementation, the submission year automatically gets closed after the deadline has passed and status criterion has been removed. This show evidence of working in an iterative approach, to make sure the requirement is achieved at the highest standard as possible.

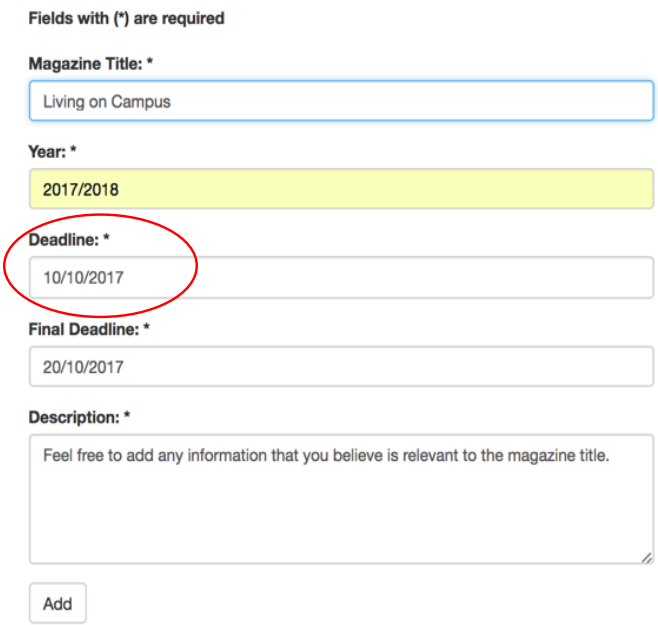

Figure 5. Create a submission year

To enhance the system further, a functionality to enable the administrator to upload the final publication in PDF format along with an image, which is shown on the homepage (index.php) could have been implemented. Currently, the publications on the homepage are static and requires a basic HTML link to redirect the user to the publication shown in a PDF format. In the meanwhile, a documentation could be produced to show how this could be achieved.

Figure 6. Published articles displayed on the homepage

user_id	username	email	password
9	john	john@gmail.com	527bd5b5d689e2c32ae974c6229ff785
10	student	student@gmail.com	cd73502828457d15655bbd7a63fb0bc8
11	admin	admin@gmail.com	21232f297a57a5a743894a0e4a801fc3
12	moses	moses@gmail.com	52bd43d37ed62eb4c226e31841bc03dc
13	mo	mo@gmail.com	27c9d5187cd283f8d160ec1ed2b5ac89
14	ali	ali@gmail.com	86318e52f5ed4801abe1d13d509443de

In accordance to the main requirement, which was to develop a secure-role based application, the system has one login form as opposed to multiple login form for each user role. Enabling one login form enhanced security and avoids confusion. To enhance security even further, passwords are encrypted using MD5 in the database. Therefore, if data theft does occur, the hacker will not be able to view the password as it will be encrypted.

Figure 7. Login webpage with pass encryption (MD5)

An important requirement that was set was to create a user-friendly web application by considering compatibility issues, therefore the web application that was developed was responsive allowing the user to navigate across webpages from smaller screens such as tablets and mobile mobiles. Creating a responsive web design will help develop user retention. As a result, this requirement was successfully achieved.

As a programmer, it was imperative to keep in mind Nielsen's Heuristics as a guideline to develop an effective web application based upon error prevention, user interface and navigation considerations. One heuristic is help users recognize, diagnose, and recover from errors. *"Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution."* (Nielsen, 1994). Therefore, server-side error preventions have been put into place to disallow the user from inserting incorrect data. Figure 8 shows the error message shown to the user when attempting to upload an incorrect file type.

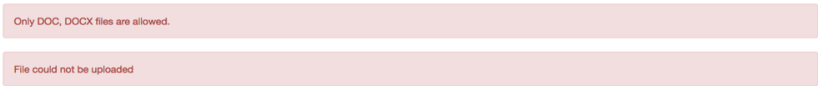

Figure 8. Error message when attempting to upload an incorrect file type

In summary, it is evident that the requirements have been met but also at a high standard for easy use for all user roles, type of users ranging from a marketing coordinators first management system to accessibility considerations for disabled users. As mentioned, the requirements gathered were developed in an iterative and incremental fashion, to make sure the best solution is put forward to achieve the required result.

2.2 Process

Due to the scope of the project, the methodology that was used during the process of the project was scrum which consisted of a team of five who were assigned individual roles in achieving the result that was required. A reason as to why this agile methodology worked is because of collaboration through effective team working principles and methods. One of the method that was used to make sure the communication between each team members is effective, was regular meetings, which consisted of reviewing each team member performance in the group. Meeting minutes also allows the team to schedule appropriately the tasks that should be completed before the next meeting, ensuring the product is delivered before the deadline. *(Refer to 'Meeting Minutes' folder in repository)*

Another method that was used to make sure the product is developed appropriately is through the implementation of a user story which shows all the features that should be implemented into the product before it is released. The user stories were produced based on the requirements per user role. It allowed the programmer (myself) fully understand the functionalities for each user role and the benefits. In some instances, the user stories were too large to complete in one iteration, therefore it was split into multiple smaller user stories before it was worked on.

The method worked effectively well, as it allowed the team to understand the project requirements further and how each functionality should work based on the user role. Sprints were also used which were short so results are delivered and ready to be tested as soon as possible. This allowed space for new features to be implemented and corrections to be made frequently based on the client's feedback. Thus, it sped up the process of fixing errors and bugs found in the system and time for development of new features. The list of tasks was identified by the scrum team which was to be developed in the sprint. This consisted of several user stories, and separate tasks that is needed to complete each user story. The method worked well, as it allowed me to understand the correct approach to take when completing each user story as opposed to not planning correctly.

To manage tasks effectively, a repository was created which included all the artefacts such as the burn down chart, meeting minutes and the sprint chart. The repository helped make sure the team members are performing to the standard that is required, by completing individual tasks and also keeping up to date with the development of the website. The artefacts in the repository were constantly updated such as the development files and product backlog to help notify the completion of each iteration.

The use case diagram *(Refer to 'Design' folder in repository)* that was developed helped the me understand the functionalities each user role has and if the same functionalities are the same for multiple user roles. It allowed the me to visualise how the system will perform by capturing the requirements from the user stories and helped visualise the things that could go wrong before developing the website. The use-case diagram helped aid the success of this project during the planning process before the development began.

Although there were several advantages using scrum during the process of the project, some phases did not work well. The chances of product failure are high if team members on the project are not committed. During the beginning stages in this project, there were times were there was a lack of

cooperation and commitment between some individuals on the project. However, to pick up on this negative characteristic, daily meetings were scheduled where the scrum master helped guide the team to work efficiently by effective leadership skills and through motivation and praise. Although there were minor disadvantages during the process which were fixed upon, using the scrum methodology helped aid the success of the project due to its methods, techniques and collaboration through effective team working principles.

2.3 Team evaluation

FACTOR	WEIGHT	Afsana	Moses	Ali	Mohammed	Alfeeque
Knowledge (Individual Role)	10	8	9	9	9	9
Logical Thinking	10	9	10	10	10	10
Initiative	10	10	10	9	10	9
Communication	10	9	9	9	9	9
Ideas	10	8	8	8	9	8
Problem Solving	10	9	9	9	10	9
Commitment	10	8	10	10	10	10
TOTAL	70	8.7	9.3	9.1	9.6	9.1

PERSON	TOTAL
Afsana	8.7
Moses	9.3
Ali	9.1
Mohammed	9.6
Alfeeque	9.1

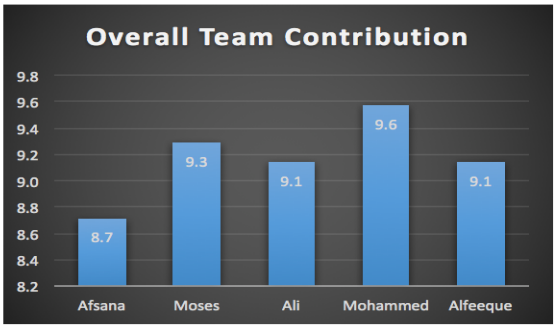

Figure 9. Team contribution scoring model using Excel spreadsheet

Name	Scoring	Justification
Mohammed (Programmer)	9.6	I believe my contribution towards the project was higher than all the other team members due to many of the factors shown in the spreadsheet. The knowledge I had in my individual role was excellent, as I've worked on PHP applications in the past however there were one or two functionalities such as pagination I was struggling on, which required further self-study to be able to develop this functionality on time. My commitment towards the project was extremely high, as I enjoyed developing this website and co-operated well regardless of whether I had a programming problem. I contributed many ideas such as change password facility, create a new submission year and many more idea that are used in website for the benefit of the client however, I could have communicated with the team members better so they are aware of the functionalities that have been implemented.
Moses (Database Engineer)	9.3	Moses contributed hugely to the success of the project due to his extensive knowledge in database design. The conceptual and logical diagram were created perfectly, however there very minor issues that were fixed after consultation. Moses was fully committed to the project, by attending all meetings and making progress every week. However, Moses could have shared more ideas during project cycle or come up with suitable solutions when issues occurred.

Ali (Web Designer)	9.1	Due to the experience in creating designs, Ali was extremely knowledgeable in his role and helped myself (programmer) fully understand how each webpage should look for enhanced user experience and easy navigation. He was also fully committed during the process of the project and developed the web designs at a high standard and on time. To improve, Ali could have also shared more ideas as a web designer.
Alfeeque (Tester/Scrum Master)	9.1	As a scrum master, Alfeeque made sure every individual team member fully understands the tasks that should be completed. He also praised and motivated individuals to perform at the standard that is required and this contributed to the success of the project. To improve, as a tester, he could have performed more complex tests such as using different characters on input forms (*,$,£,#) to see how the system responds. SQL injections could have also been used to see what occurs as security is a big concern in any web application. Nonetheless, Alfeeque showed great character, effort and understanding in both roles which contributed to the success of the project.
Afsana (Product Owner)	8.7	Communication between Afsana and the client was at a good standard to identify the requirements that need to be delivered which was further passed onto the other team members. Her initiative was also good as she could assess things differently than other team members. However, as a product owner she lacked some knowledge in her role such as developing graphs. She could have also been committed more in making sure the website is developed to the requirements that she obtained from the clients are met. Some tasks were also not completed on time.

2.4 Individual contribution

I believe my contribution to this project was at the highest level as possible, by contributing hugely on the development of the web application. Before the start of the project, I took on the responsibility of being the programmer of the team due to my experience in create multiple web application that were a success. I fully understood the scope of the project; what the application provides and what is required by the client.

As a programmer, it was extremely important that I communicate and work with all members of the team which included the web designer, database engineer, tester and the product owner. I believe I was one of the team members who had on-going communication with all the team members on a regular basis to get the project developed at the standard that was required. I contributed many ideas during the process of the project, for instance allowing the administrator to create a new submission year that's based on a specific topic, automatically closing submission year depending on the deadline day, changing password and many more. All these were not required requirements, although I thought it would be advantageous to have to make it easy for admins and create a better user experience.

Firstly, I worked alongside the web designer to go through all the designs that was developed using Axure to see whether this is achievable to be coded in HTML and CSS within the time frame *(Refer to 'Design' folder in repository)*. The design wasn't as complex, as the web designer went for easy use rather than complexity. The development phase of the web design was achieved exactly to the wireframes that were produced by the help of a relevant framework, bootstrap. After developing the web design on HTML, there were some webpages that needed to be improved, therefore there was

constant on-going communication between myself (programmer) and the web designer to achieve the best possible design.

Databases are extremely important in any application as it provides the structure of the application. Therefore, it was imperative to make sure the database implemented is sufficient to hold data whilst considering the correct data types and referential integrity *(Refer to 'Database Diagram' in repository)*. I worked with the database engineer to make sure the conceptual and the logical diagram is correct before implementing SQL codes to implement the database on MYSQL. The tester on the other hand made sure the web application is error and bug free, although in some instances the test that were taken failed therefore, it was my responsibility in making sure they are *fixed* before release *(Refer to 'Testing' folder in repository)*.

The presentation that was conducted, explained the benefits of the system, testing that was done and the reasons to buy the website. The section I took on was to demonstrate some of the functionalities on the website and explain the approach we took for some of them by considering user experience and the ease of navigation. I took on this part because I fully understood what is required and the approach we took in developing the application. Due to this, I also took on the responsibility of producing a panopto, showcasing the web application and it's functionalities.

The final product that was developed was solely by myself but through the help from all team members. All the functionalities, web design implementation using HTML, CSS, PHP, Bootstrap and the database implementation using SQL was produced by myself although the final product could have not been implemented without the assistance from all the team members. Although this was a team based project, I also had to work individually as a sole programmer in the project. Due to my experience in create web applications, all the functionalities were developed at an extremely high standard but also on time. There were one or two instances where I was challenged upon, for example the implementation of a pagination where I struggled to think of the correct approach to take. However, due to my motivational drive to learn and succeed, I managed to develop pagination after reading books on PHP and using online resources to research upon the issues faced.

A personal factor that worked well was my individual determination to get the website functioning to my highest ability by conducting independent learning and using my skills that I have as an advantage to meet the course specification that was required. In terms of implementation of the website, majority of the functionalities were produced without many difficulties. However, if there were problem, I would consult with the team members and try to come up with a solution and in terms of programming help, I used online resources and books to find solutions to the problems faced. This enhanced my problem solving skills.

3. Lesson learnt

In terms of good working practices, my time management skills were at a high standard. The project required many functionalities to be implemented within a short time, therefore to make sure the website is developed before the deadline, prioritising the work load into sprints helped aid the success of the project. This could have not been achieved if my time management skills were poor, ultimately result in project failure.

Working in a scrum team required multiple personal qualities that will help work effectively in a team. One personal skill that was used frequently, was my communication skills where there was frequent communication between all the team members to make sure the project is developed appropriately to the requirements specified. This also consisted of being able to listen effectively, by taking on feedback from other team members that helped develop the functionalities at a higher standard. As a programmer, it was important that I develop quality code that can be readable by other developers. The coding that was produced is also structured well, for instance the function.php *(Refer to 'Project File' in repository)* has all the PHP server side codes used to develop the functionalities such as login and registering new users. Codes have also been commented out to make it understandable for non-technical users. The development required extensive problem solving skills, in order to make sure the correct approach is taken. A problem that was faced but later solved was URL routing, where the user role should be limited to what they are able to view. This was achieved by simply putting the user role in a session and using an if statement to show the content or redirect the user to the error page. Without having good problem solving skills, implementation of the functionalities would be at a poor standard. Figure … shows a code snippet of how URL routing was accomplished.

```
if (isset($_SESSION['username']) && isset($_SESSION['role']) && $_SESSION['role'] == 'Admin') {

} else {

    header('Location: ./error.html');
}
```

Figure 10. URL Routing Code snippet

After the completion of the project, there were many things that could be improved whilst working in a scrum team but also extra functionalities on the website that can improve the web application. It may be ideal to have another programmer in the team, to minimise the extensive work load for the project also, pair programming could be used if there was another programmer in the team, which will help reduce the errors made, develop team cohesion and help develop structured quality code.

If there project was to be done again, I would make sure the tests are done as soon as possible after each iteration to make sure the product is error and bug free. This also allows myself to focus on making one improvement at a time, as opposed to working on multiple fixes after the functionalities have been tested. Having another developer in the team can help fix the failed tests quicker.

A full stack developer/programmer should also consider using more client side programming, for example JavaScript which can enhance user experience. JavaScript could have been used in the future for error preventions and is proven to be relatively fast to the end user as the code is executed on the

client's computer. In terms of future development, as the website is in co-operation with a University, it may be ideal to integrate the primary University user database table with the current database used for Swift magazine. This would enhance referential integrity as a student or a marketing co-ordinator will not have different username and password for the University account and Swift. Thus, it will reduce the administrator's task in registering every University student to Swift moreover, a registration functionality will not be needed.

4. Conclusion

To conclude, the success of the project was due to use of the agile methodology scrum. It provides a depth for team collaboration, team morale and better quality of delivery. Each member in the team helped develop the web application, and the role was provided dependant on individual strengths and qualities to get the website functioning based on the requirements set. The product was developed in an iterative fashion to make sure each user story is complete, to the standard that is required by the client but also those who will be using the system daily. The requirements gathered were clear although there were some requirements that needed to be expanded on, therefore communication between the scrum team and the client was on-going to further develop the understanding of each requirement before developing a user story, product backlog and a sprint backlog. Scrum was the correct methodology to use as is clearly showcased the advantages for the scale of this project.

5. References

Nielsen, J. (1994). 10 Heuristics for User Interface Design: Article by Jakob Nielsen. [online] Nngroup.com. Available at: https://www.nngroup.com/articles/ten-usability-heuristics/ [Accessed 11 Apr. 2017].

6. Bibliography

Davis, M. and Phillips, J. (2008). Learning PHP and MySQL. 1st ed. Sebastopol: O'Reilly Media, Inc.